

The sharp Frobenius-Rieffel constants from the Pimsner-Popa finite-dimensional index formula

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Status. Literature-implied answer to the full sharp-constant problem, likely valid, subject to human review. The supporting authors did not identify their result as an answer to the Frobenius-Rieffel question.

1 Source question

For a finite-dimensional unital C^* -algebra $\mathcal{A}$, a unital C^* -subalgebra $\mathcal{B} \subseteq \mathcal{A}$, a faithful tracial state τ , and the unique τ -preserving conditional expectation $E : \mathcal{A} \rightarrow \mathcal{B}$, Aguilar, Garcia, and Kim define

$$\|x\|_E = \|E(x^*x)\|_{\text{op}}^{1/2}.$$

They ask for the sharp constant

$$\kappa(E) = \inf_{x \neq 0} \frac{\|x\|_E}{\|x\|_{\text{op}}}. \quad (1)$$

The question appears immediately before Table 1 on arXiv PDF page 14; the table of conjectured constants is on page 15 [1]. The paper treats arbitrary finite-dimensional $\mathcal{A}$, $\mathcal{B}$, and faithful traces, so “find the sharp constants” is naturally read in this full generality.

2 The overlooked identification

Write

$$\mathcal{A} = \bigoplus_{k=1}^K M_{d_k}(\mathbb{C}), \quad \mathcal{B} \cong \bigoplus_{j=1}^J M_{n_j}(\mathbb{C}).$$

After unitary conjugacy, the representation of $\mathcal{B}$ in the k th summand is

$$(z_j)_j \mapsto \bigoplus_{j=1}^J (I_{m_{kj}} \otimes z_j), \quad d_k = \sum_j m_{kj} n_j, \quad (2)$$

where $m_{kj} \geq 0$ and every column is nonzero. If

$$\tau((x_k)_k) = \sum_k q_k \text{Tr}(x_k), \quad q_k = \frac{v_k}{d_k}, \quad v_k > 0, \quad \sum_k v_k = 1,$$

put

$$\omega_j = \sum_{\ell=1}^K q_\ell m_{\ell j}, \quad b_{kj} = \min\{m_{kj}, n_j\},$$

and

$$D_k = \sum_{j: m_{kj} > 0} \frac{b_{kj} \omega_j}{q_k}, \quad D = \max_k D_k. \quad (3)$$

Theorem 1 (sharp constant). *For every finite-dimensional inclusion and faithful trace above,*

$$\boxed{\kappa(E) = D^{-1/2}}. \quad (4)$$

Moreover, D is exactly the reciprocal of the finite-dimensional Pimsner-Popa order constant computed in Theorem 6.1 of [2].

Identification with Pimsner-Popa. Pimsner and Popa write a_{kj} for the inclusion multiplicities, q_k for the trace of a minimal projection in the k th ambient factor, and ω_j for the trace of a minimal projection in the j th subalgebra factor. Their Theorem 6.1, on printed page 93 (supporting PDF page 38), states

$$\lambda(\mathcal{A}, \mathcal{B})^{-1} = \max_k \sum_j \min\{a_{kj}, n_j\} \frac{\omega_j}{q_k}. \quad (5)$$

Thus (3) is precisely their formula. Since $E(c) \geq D^{-1}c$ for $c \geq 0$, it immediately gives $\|x\|_E \geq D^{-1/2}\|x\|_{\text{op}}$. The following elementary equality case shows that this lower constant is sharp for the norm problem, not merely for the stronger order inequality.

Equality witness. Choose k_0 with $D_{k_0} = D$. In the decomposition (2), choose orthonormal vectors $e_{jr} \in \mathbb{C}^{m_{k_0j}}$ and $f_{jr} \in \mathbb{C}^{n_j}$ for $1 \leq r \leq b_{k_0j}$, and set

$$x_j = \sqrt{\frac{\omega_j}{q_{k_0}D}} \sum_{r=1}^{b_{k_0j}} e_{jr} \otimes f_{jr}, \quad x = \bigoplus_j x_j.$$

Equation (3) gives $\|x\| = 1$. Let $p = xx^*$, supported in the k_0 th summand of $\mathcal{A}$. The j th abstract component of $E(p)$ is

$$\frac{q_{k_0}}{\omega_j} \text{Tr}_{\mathbb{C}^{m_{k_0j}}} (x_j x_j^*) = \frac{1}{D} \sum_{r=1}^{b_{k_0j}} f_{jr} f_{jr}^*.$$

Consequently $\|E(p)\|_{\text{op}} = 1/D$. Since $p = p^*p$ and $\|p\|_{\text{op}} = 1$, equality holds in (4).

For completeness, the lower bound can also be seen without index theory. A positive norm-one $c \in \mathcal{A}$ dominates a rank-one projection in a summand where its norm is attained. For such a projection, write t_j for the trace of its j th reduced density matrix. Its rank is at most b_{kj} , so if $R = \|E(p)\|_{\text{op}}$ then

$$t_j \leq R \frac{b_{kj}\omega_j}{q_k}, \quad 1 = \sum_j t_j \leq RD_k \leq RD.$$

Hence $R \geq 1/D$, and positivity of E completes the same proof.

3 Matrix-algebra specialization and a corrected guess

For $\mathcal{A} = M_n$ and $\mathcal{B} = \bigoplus_j (I_{m_j} \otimes M_{n_j})$, formula (3) reduces to

$$D = \sum_j m_j \min\{m_j, n_j\}. \quad (6)$$

This is also the specialization recorded as the Pimsner-Popa index in modern quantum Markov literature [3, formula (14)].

Formula (6) agrees with every numerical guess in Table 1 of [1] except one. For

$$\mathcal{B}_{2^2,1}^5 = \{\text{diag}(A, A, \lambda) : A \in M_2, \lambda \in \mathbb{C}\},$$

it gives $D = 2 \min(2, 2) + 1 = 5$, hence $\kappa = 1/\sqrt{5}$, not the guessed $1/2$. Explicitly,

$$x = (\sqrt{2/5}, 0, 0, \sqrt{2/5}, \sqrt{1/5})^T, \quad p = xx^*,$$

satisfies $E(p) = I_5/5$.

4 Provenance and scope

This is classified as a literature-implied full answer, not as a new theorem: the decisive multiplicity-and-trace formula is Pimsner and Popa’s 1986 Theorem 6.1. The bridge to the Frobenius-Rieffel norm and its explicit rank-one equality case do not appear to have been noted by the source authors. Indeed, their March 2026 follow-up still says that the sharp constants are unknown [4]. Bounded searches used the exact source title, author names, “sharp Frobenius-Rieffel constant”, “Pimsner-Popa index”, and finite-dimensional multiplicity formulas. No separate explicit resolution in Frobenius-Rieffel terminology was located.

Human review focus. Check the translation of minimal-projection trace weights in (5) to q_k, ω_j , and verify that the unitary standard form (2) covers linked simple components across distinct ambient summands. The rank-one equality calculation is otherwise finite-dimensional linear algebra.

References

- [1] K. Aguilar, S. R. Garcia, and E. Kim, *Frobenius-Rieffel norms on finite-dimensional C^* -algebras*, Operators and Matrices 16 (2022), 733–758; arXiv:2112.13164.
- [2] M. Pimsner and S. Popa, *Entropy and index for subfactors*, Ann. Sci. Ecole Norm. Sup. (4) 19 (1986), 57–106.
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- [4] K. Aguilar, S. R. Garcia, E. Kim, and F. Latremoliere, *The strongly Leibniz property and the Gromov-Hausdorff propinquity*, arXiv:2301.05692, March 2026 revision.